LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Equation Sheet: Every Equation You Need (Class 10)

Every equation on this sheet has been checked atom by atom. Cover the middle column and try to write each equation yourself, then check. Add the states (s), (l), (g), (aq) when a question asks for them.

Ch 1: Chemical Reactions and Equations

Reaction	Balanced equation	Remember
Burning magnesium ribbon	$2Mg + O_2 \rightarrow 2MgO$	Dazzling white flame; white powder (combination)
Zinc + dilute sulphuric acid	$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$	Hydrogen gas; tube gets warm
Lead nitrate + potassium iodide	$Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3$	Yellow precipitate (double displacement)
Quicklime + water	$CaO + H_2O \rightarrow Ca(OH)_2$	Slaked lime; lots of heat (exothermic combination)
Whitewash shine	$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$	Thin shiny layer of $CaCO_3$ after 2–3 days
Burning coal / hydrogen	$C + O_2 \rightarrow CO_2$ · $2H_2 + O_2 \rightarrow 2H_2O$	Combination
Burning natural gas	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	Exothermic
Respiration	$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy	Exothermic
Heating ferrous sulphate	$2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$	Green → reddish-brown; smell of burning sulphur
Heating limestone	$CaCO_3 \rightarrow CaO + CO_2$	Thermal decomposition; used to make cement
Heating lead nitrate	$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$	Brown fumes of NO_2
Electrolysis of water	$2H_2O \rightarrow 2H_2 + O_2$	H_2 volume is double the O_2 volume
Silver chloride in sunlight	$2AgCl \rightarrow 2Ag + Cl_2$	White → grey; black-and-white photography
Silver bromide in sunlight	$2AgBr \rightarrow 2Ag + Br_2$	Photolytic decomposition
Iron nail in copper sulphate	$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$	Blue → light green; brown coat on the nail
Zinc / lead displacing copper	$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$ · $Pb + CuCl_2 \rightarrow PbCl_2 + Cu$	Displacement
Sodium sulphate + barium chloride	$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$	White precipitate of $BaSO_4$
Heating copper; then hydrogen over it	$2Cu + O_2 \rightarrow 2CuO$ · $CuO + H_2 \rightarrow Cu + H_2O$	Brown → black → brown (redox)
Zinc oxide + carbon	$ZnO + C \rightarrow Zn + CO$	C oxidised, ZnO reduced
Manganese dioxide + HCl	$MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$	HCl oxidised, MnO_2 reduced
Iron + steam	$3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$	Classic balancing question

Ch 2: Acids, Bases and Salts

Reaction	Balanced equation	Remember
Zinc + sodium hydroxide	$2NaOH + Zn \rightarrow Na_2ZnO_2 + H_2$	Sodium zincate; not all metals do this
Carbonate + acid	$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$	CO_2 turns lime water milky
Hydrogencarbonate + acid	$NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$	Brisk effervescence
Lime water test	$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$	Milky
Excess CO_2	$CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO_3)_2$	Milkiness disappears (soluble)
Neutralisation	$NaOH + HCl \rightarrow NaCl + H_2O$	Acid + base → salt + water
Metal oxide + acid	$CuO + 2HCl \rightarrow CuCl_2 + H_2O$	Solution turns blue-green
Chlor-alkali process	$2NaCl + 2H_2O \rightarrow 2NaOH + Cl_2 + H_2$	Cl_2 at anode, H_2 at cathode, NaOH near cathode
Bleaching powder	$Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$	Chlorine over dry slaked lime

Making baking soda	$NaCl + H_2O + CO_2 + NH_3 \rightarrow NH_4Cl + NaHCO_3$	Solvay-type process
Heating baking soda	$2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$	CO_2 makes cakes rise
Washing soda	$Na_2CO_3 + 10H_2O \rightarrow Na_2CO_3 \cdot 10H_2O$	Recrystallisation
Heating blue vitriol	$CuSO_4 \cdot 5H_2O \rightarrow CuSO_4 + 5H_2O$	Blue → white; water brings blue back

Plaster of Paris: gypsum $CaSO_4 \cdot 2H_2O$ heated at 373 K gives $CaSO_4 \cdot \frac{1}{2}H_2O + 1\frac{1}{2}H_2O$. Mixed with water, it sets back into gypsum (a hard mass).

Ch 3: Metals and Non-metals

Reaction	Balanced equation	Remember
Sodium + oxygen	$4Na + O_2 \rightarrow 2Na_2O$	Basic oxide; Na is stored in kerosene
Sodium / potassium + cold water	$2Na + 2H_2O \rightarrow 2NaOH + H_2$ · $2K + 2H_2O \rightarrow 2KOH + H_2$	Violent; H_2 may catch fire
Calcium + water	$Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$	Calcium floats (H_2 bubbles stick to it)
Magnesium + hot water	$Mg + 2H_2O \rightarrow Mg(OH)_2 + H_2$	Also floats
Aluminium / iron + steam	$2Al + 3H_2O \rightarrow Al_2O_3 + 3H_2$ · $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$	Metal oxide + hydrogen
Amphoteric oxide	$Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$ · $Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$	Reacts with acid AND base
Metals + dilute HCl	$Fe + 2HCl \rightarrow FeCl_2 + H_2$ · $Mg + 2HCl \rightarrow MgCl_2 + H_2$	Only metals above H
Mercury from cinnabar	$2HgS + 3O_2 \rightarrow 2HgO + 2SO_2$ · $2HgO \rightarrow 2Hg + O_2$	Low reactivity: heat alone
Copper from Cu_2S	$2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$ · $2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$	Self-reduction
Zinc: roasting / calcination	$2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$ · $ZnCO_3 \rightarrow ZnO + CO_2$	Sulphide: excess air · carbonate: limited air
Zinc: reduction	$ZnO + C \rightarrow Zn + CO$	Carbon (coke) as reducing agent
Aluminium as reducing agent	$3MnO_2 + 4Al \rightarrow 3Mn + 2Al_2O_3 + \text{heat}$	Highly exothermic
Thermite reaction	$Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3 + \text{heat}$	Joins railway tracks

Electrolysis of molten NaCl: at the cathode $Na^+ + e^- \rightarrow Na$; at the anode $2Cl^- \rightarrow Cl_2 + 2e^-$. **Ionic bonds:** $Na \rightarrow Na^+ + e^-$; $Cl + e^- \rightarrow Cl^-$; $Mg \rightarrow Mg^{2+} + 2e^-$.

Ch 4: Carbon and its Compounds

Reaction	Balanced equation	Remember
Combustion	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ · $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$	Heat and light
Hydrogenation (addition)	$C_2H_4 + H_2 \rightarrow C_2H_6$	Ni or Pd catalyst; oil → vegetable ghee
Substitution	$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$	Needs sunlight
Ethanol + sodium	$2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2$	Sodium ethoxide + hydrogen
Dehydration of ethanol	$C_2H_5OH \rightarrow C_2H_4 + H_2O$	Hot conc. H_2SO_4, 443 K
Esterification	$CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$	Acid catalyst; sweet smell
Saponification	$CH_3COOC_2H_5 + NaOH \rightarrow C_2H_5OH + CH_3COONa$	Used to make soap
Ethanoic acid + base	$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$	Sodium acetate
Ethanoic acid + carbonates	$2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$ · $CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + H_2O + CO_2$	Effervescence: tells acid from alcohol

Oxidation of ethanol (written with the oxidising agent over the arrow, not balanced): $CH_3CH_2OH \rightarrow CH_3COOH$, using alkaline $KMnO_4$ + heat or acidified $K_2Cr_2O_7$.

Biology & environment

Process	Balanced equation	Remember

Photosynthesis	$6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$	Sunlight + chlorophyll
Ozone formation	$O_2 \rightarrow O + O \cdot O + O_2 \rightarrow O_3$	UV splits O_2

Respiration word paths (no equations needed): glucose → pyruvate (cytoplasm) → ethanol + CO_2 (yeast) / lactic acid (muscles) / CO_2 + water (mitochondria).